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Week 2 and 3

GEITE01X – Living in the IT Era

Science, Culture, & Ethics in IT

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LEARNING OUTCOMES

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At the end of the lesson, the students should be able to:

1. Connect science and technology to the real world
2. Identify and engage in a variety of cultural media, artistic, and others
3. Describe the importance of ethics and professional conduct in the workplace.

OFFICE 365 AND WINDOWS MODULES

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1. Create authentic assessments with Microsoft Forms
2. Get Started with OneNote
3. Structure Teams through channels, tabs, files, and apps
4. Organize content, create assignments, and assess learners' understanding in Teams
5. Empower school leaders and tech-savvy educators with Windows 11 and Windows 11 SE: Course 101
6. Engage teachers and students with Windows 11 and Windows 11 SE: Course 201

ABOUT SCIENCE AND TECHNOLOGY

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What is Science?

- Science fundamentally is the systematic study of the structure and behavior of the natural and physical world through observations and experiments.



DIFFERENT FACET OF SCIENCE

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- Science is both a body of knowledge and a process
- Science is exciting
- Science is useful
- Science is ongoing
- Science is a global human endeavor



TECHNOLOGY

- **Technology** *(which is basically derived from the Greek word '**technologia**')* is an art, skill or ability that is used to create and develop products and acquire knowledge.



CONTRIBUTION OF SCIENCE TO TECHNOLOGY



Science as a direct source of new technological ideas



Science as a source of engineering

CONTRIBUTION OF SCIENCE TO TECHNOLOGY



Technology as a source of scientific challenges



Instrumentation and measurement techniques

NATURE OF SCIENCE AND TECHNOLOGY

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Technical Knowledge - It can be defined in the simplest terms as ***'know-how'.***

Understanding of the attributes or elements - It means knowledge and understanding of the intelligence of workers, quality of products, value of a firm, effectiveness of the market, etc.



NATURE OF SCIENCE AND TECHNOLOGY

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Nature of Science

- **Science** is the study of the natural world.
- It aims to **discover facts, principles, and laws** about how things work.
- Science uses **observation, experimentation, and evidence** to explain phenomena.
- Its main goal is to **gain knowledge and understanding.**

Example: Scientists study how plants grow, why the sky is blue, or how diseases spread.

Nature of Technology

- **Technology** is the practical application of scientific knowledge.
- It focuses on **creating tools, systems, or processes** to solve problems and make life easier.
- Technology is **human-made** and constantly evolving.
- Its main goal is to **apply knowledge for useful purposes.**

Example: Engineers use science about electricity to create light bulbs, computers, or mobile phones.

NATURE OF SCIENCE AND TECHNOLOGY

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Science and Technology Together

Science provides knowledge	Technology applies it.
Science provides knowledge	Science uses them to make new discoveries.

They are interconnected:
Science drives technology, and technology creates new challenges for science.



Science:
knowing why things happen.

Technology:
using that knowledge to make things work.

ABOUT CULTURE AND TECHNOLOGY

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What is Culture?

- Culture is an umbrella term which encompasses the social behavior and norms found in human societies, as well as the knowledge, beliefs, arts, laws, customs, capabilities, and habits of the individuals in these groups.



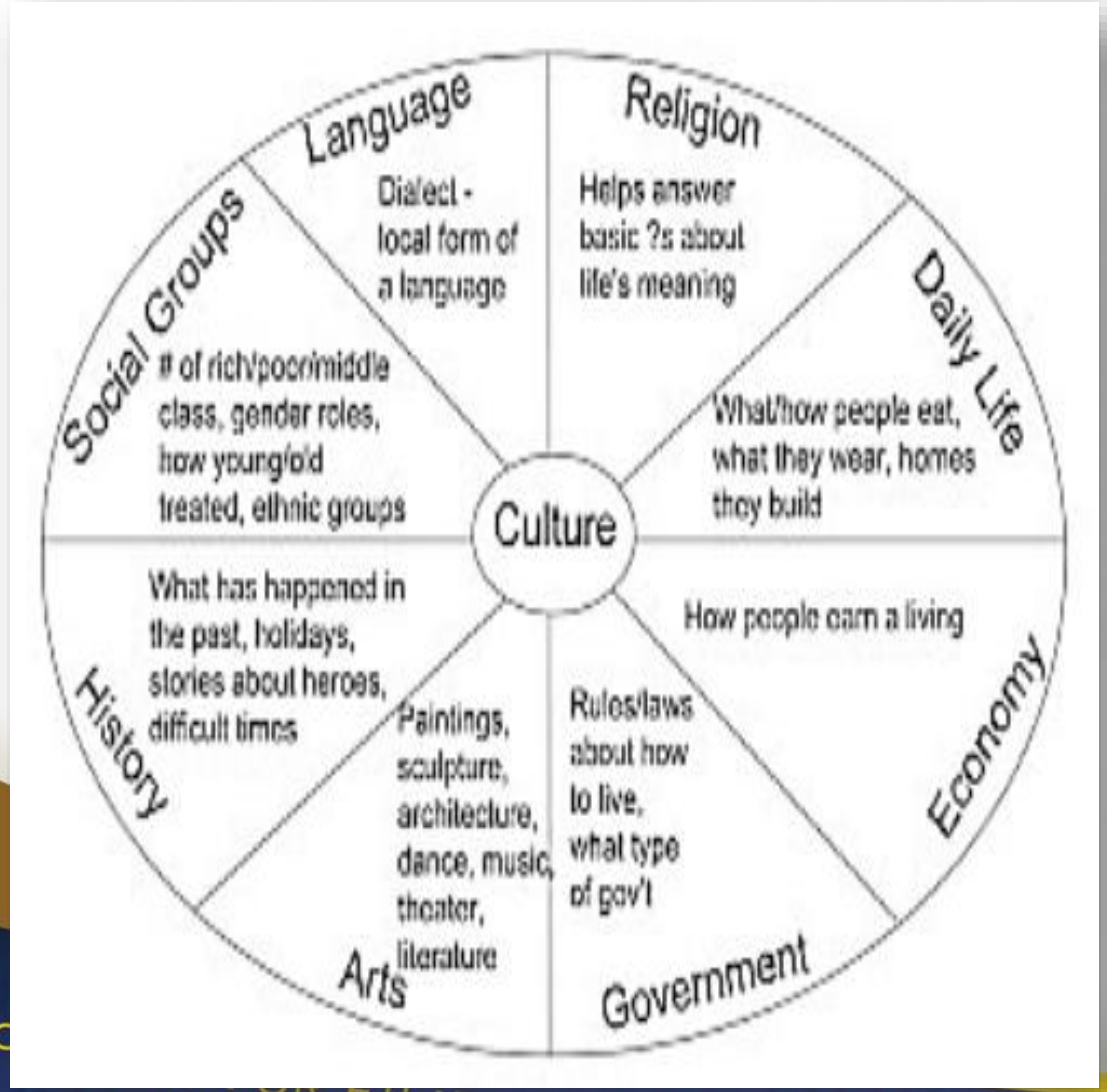
ABOUT CULTURE AND TECHNOLOGY

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Definition from different sociologists.

- Taylor: “Culture is the complex whole which includes knowledge, belief, art, morals, law, customs and habits, and any capabilities acquired by man as a member of society”
- Linton: “Culture is social heredity, which is transmitted from one generation to another with the accumulation of individual experiences”
- John Beattee: “Culture is the way of life which is transmitted from generation to generation

ABOUT CULTURE AND TECHNOLOGY



The 8 Elements of Culture

1. Religion
2. Daily Life
3. Economy
4. Government
5. Arts
6. History
7. Social Group
8. Language

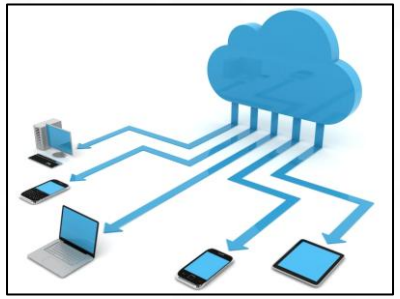
TECHNOLOGY SHAPES CULTURE

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- The technology landscape for businesses has changed beyond recognition over the past 50 years. Here are 5 ways that technology has changed the way that we work.



Mobility



The Cloud



VR &AR Innovation



Customer Data



Social Impact

CONNECTION OF CULTURE TO TECHNOLOGY

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- **Culture influences technology:** The needs, values, and traditions of a society guide the kind of technologies people create.

  Example: In agricultural societies, tools like plows and irrigation systems were invented because farming was central to their culture.

- **Technology influences culture:** Once created, technology changes how people live, think, and interact.

  Example: Social media reshaped communication, relationships, and even cultural trends worldwide.

IN SIMPLE TERMS

Culture inspires technology (what people create).
Technology shapes culture (how people live and think).
They are interconnected — one always affects the other.

ABOUT ETHICS OF INFORMATION TECHNOLOGY

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Every society forms a set of rules that establishes the boundaries of generally accepted behavior. These rules are often expressed in statements about how people should behave, and the individual rules fit together to form the moral code by which a society lives.

What is Ethics?

- **Ethics** is a set of beliefs about right and wrong behavior within a society. Ethical behavior conforms to generally accepted norms many of which are almost universal.

ABOUT ETHICS OF INFORMATION TECHNOLOGY

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Ethics of Information Technology is the study of what is **right and wrong** in the use of computers, digital systems, and information. It guides how people should use technology in a fair, safe, and responsible way.



ABOUT ETHICS OF INFORMATION TECHNOLOGY

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Key Areas of IT Ethics

- Privacy
- Security
- Intellectual Property
- Digital Responsibility
- Professional Conduct

Why IT Ethics is Important

- Builds **trust** in digital systems.
- Protects **rights and freedoms** of individuals.
- Ensures technology is used for **good, not harm**.
- Helps professionals make **fair and just decisions** when facing ethical dilemmas.

ABOUT ETHICS OF INFORMATION TECHNOLOGY

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MORALITY

- The term **morality** refers to social conventions about right and wrong that are so widely shared that they become the basis for an established consensus.
- **Morals** are the **principles of right and wrong** that guide a person's behavior.
- They come from **society, culture, religion, or personal beliefs.**
- They tell us what is considered **acceptable or unacceptable.**
  - Example: Being honest, respecting elders, and not stealing are considered moral actions.

ABOUT ETHICS OF INFORMATION TECHNOLOGY

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VIRTUE

- A **virtue** is a habit that inclines people to do what is acceptable, and a vice is a habit of unacceptable behavior.
- Virtues are the **good qualities or habits of character** that make a person morally good.
- They are **practiced consistently** until they become part of one's personality.
- Virtues lead to living a **good and meaningful life.**
  - Example: Courage, kindness, patience, humility, and generosity are virtues.

ABOUT ETHICS OF INFORMATION TECHNOLOGY

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Difference

MORALS	VIRTUES
The rules or principles (knowing right from wrong)	The practiced habits of goodness (living out those morals)

Example:
Morality tells you that **helping others is good.**

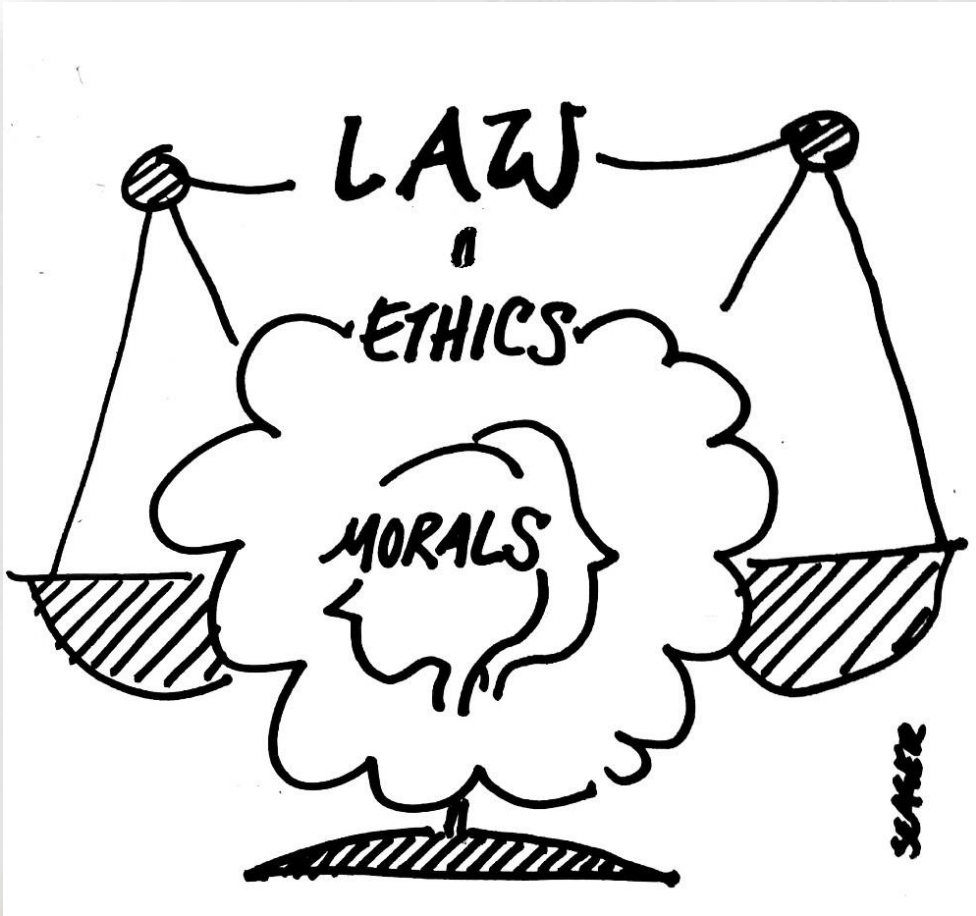
Virtue is when you actually **practice kindness** regularly.

IN SIMPLE TERMS

Morals guide actions.
Virtues shape character.

MORALS, ETHICS, AND LAWS

- **MORALS** are one's personal beliefs about right and wrong.
- **ETHICS** describes standards or codes of behavior expected of an individual by a group (nation, organization, profession) to which an individual belongs.
- **LAW** is a system of rules that tells us what we can and cannot do. Laws are enforced by a set of institutions (the police, courts, law-making bodies). Legal acts are acts that conform to the law.



MORALS, ETHICS, AND LAWS

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COMPARISON

MORALS	ETHICS	LAWS
What I believe is right or wrong (personal) .	*What society/ profession says is right or wrong (group).*	*What the government requires or forbids (official).*

Example:

Moral: I believe lying is wrong.

Ethic: As a teacher, I must be fair to all students.

Law: It is illegal to cheat on taxes.

IN SIMPLE TERMS

Morals guide personal behavior.
Ethics guide professional and social behavior.
Laws regulate legal behavior.

ETHICS IN IT

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- **ETHICS IN IT** refers to the rules, principles, and standards that guide how people **use, develop, and manage information technology** in a fair, responsible, and respectful way.
- It answers the question: *"What is the right or wrong way to use technology?"*



ETHICS IN IT

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- During the many IT breakthroughs in recent years, the importance of ethics and human values has been underemphasized with a range of consequences. Here are some examples that raise public concern about the ethical use of information technology:



Many employees have their email and Internet access monitored while at work, as employers struggle to balance their need to manage important company assets and work time with employees' desire for privacy and self-direction.



Millions of people have downloaded music and movies at no charge and in apparent violation of copyright laws at tremendous expense to the owners of those copyrights.

ETHICS IN IT

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Organizations contact millions of people worldwide through unsolicited email (spam) as an extremely low-cost marketing approach.



Students around the world have been caught downloading material from the Web and plagiarizing content for their term papers.



Web sites plant cookies or spyware on visitors' hard drives to track their online purchases and activities.



Hackers break into databases of financial and retail institutions to steal customer information, then use it to commit identity theft opening new accounts and charging purchases to unsuspecting victims.

PROFESSIONAL CODE OF ETHICS

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Purposes of a Professional Code of Ethics

- To guide professionals in making the **right decisions.**
- To build **trust** between professionals and the public.
- To maintain **fairness and honesty** in the workplace.
- To protect the **rights and safety** of clients, users, or society.
- To encourage **professional responsibility.**

PROFESSIONAL CODE OF ETHICS

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Core Principles in IT and Other Professions

- Integrity
- Confidentiality
- Competence
- Fairness
- Responsibility
- Respect for Intellectual Property
- Public Interest

Example in IT:
An IT professional must **not install pirated software**, **must not misuse client data**, and should **develop systems that are safe and reliable.**



A **Professional Code of Ethics** is like a **rulebook of good behavior** for professionals. It tells them how to act with honesty, fairness, and responsibility in their work.

PROFESSIONAL CODE OF ETHICS

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- **ETHICAL DECISION MAKING** is the process of **choosing the best action that is right, fair, and responsible** when faced with a problem or dilemma. It means thinking not only about what is legal or easy, but also about what is **morally correct** and benefits others.
  - **Ethical decision making:** Adherence to a professional code of ethics means that practitioners use a common set of core values and beliefs as a guideline for ethical decision making.
- **HIGH STANDARDS OF PRACTICE AND ETHICAL BEHAVIOR:** Adherence to a code of ethics reminds professionals of the responsibilities and duties that they may be tempted to compromise to meet the pressures of day-to-day business.

PROFESSIONAL CODE OF ETHICS

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- **TRUST** means people believe in your honesty, reliability, and ability to do your job well.
- **RESPECT** means people value you and your work because you act fairly, responsibly, and with integrity.
  - Together, they are the **foundation of a good professional reputation.**
- **Trust and respect from the general public:** Public trust is built on the expectation that a professional will behave ethically.



Professionals gain **trust** when people can rely on them, and they gain **respect** when they consistently do what is right and fair.

PROFESSIONAL CODE OF ETHICS

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- **Evaluation benchmark:** A code of ethics provides an evaluation benchmark that a professional can use as a means of self-assessment. Peers of the professional can also use the code for recognition or censure.
  - An **evaluation benchmark** is like a **measuring stick** — it tells us if something is doing well, average, or poorly compared to a set standard.

COMMON ETHICAL ISSUES FOR IT USERS

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- **SOFTWARE PIRACY**
  - Software piracy in a corporate setting can sometimes be directly traceable to IT professionals–they might allow it to happen, or they might actively engage in it.
- **INAPPROPRIATE USE OF COMPUTING RESOURCES**
  - Some employees use their computers to surf popular Web sites that have nothing to do with their jobs, participate in chat rooms, view pornographic sites, and play computer games.
- **INAPPROPRIATE SHARING OF INFORMATION**
  - Every organization stores vast amounts of information that can be classified as either private or confidential.

SUPPORTING ETHICAL PRACTICES OF IT USERS

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- Establishing Guidelines for Use of Company Software
- Defining and Limiting the Appropriate Use of IT Resources
- Structuring Information Systems to Protect Data and Information
- Installing and Maintaining a Corporate Firewall

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